

This is due STRICTLY by August 22. NO extension for this homework.

Problem 1. We study how is irreducibility related to factorization. It is simple to analyze the constant polynomials (i.e, the degree zero ones). Let $f(x) \in \mathbb{Q}[x]$ be of positive degree.

1. Show that $f(x)$ is irreducible iff $f(x)$ does not factor as a product of polynomials of (strictly) smaller degrees. Suppose $f(x) \in \mathbb{Z}[x]$. Does the above statement hold?
2. Suppose $f(x) \in \mathbb{Z}[x]$ and satisfies the requirements of the Eisenstein criterion. Show $f(x)$ is irreducible in $\mathbb{Q}[x]$.

Problem 2. We study the connection between primitiveness and reduction mod p . Let $f(x) \in \mathbb{Z}[x]$.

1. Show that $f(x)$ is primitive iff for every prime $p \in \mathbb{Z}$, there exists some coefficients of $f(x)$ that is not divisible by p .
2. Show that if $f(x)$ and $g(x)$ are primitive, then $f(x)g(x)$ is primitive by considering $\mathbb{Z}_p[x]$.

Problem 3. In this exercise, we prove the rational root test in two ways. Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$ and $a_n, a_0 \neq 0$. Suppose $r = p/q \in \mathbb{Q}$, where p, q coprime, is a rational root of $f(x)$.

1. Show that p is a divisor of a_0 and q is a divisor of a_n by consider $f(p/q) \in \mathbb{Q}$.
2. Show that $qx - p$ divides $f(x)$ and conclude that $p \mid a_0$ and $q \mid a_n$.

Problem 4. There are many times both reduction mod p and Eisenstein criterion can't directly determine the irreducibility of a polynomial. This requires other tools.

1. Let $f(x) \in \mathbb{Q}[x]$ (or in $\mathbb{Z}[x]$ resp.) be of positive degree and $r \in \mathbb{Q}$ (or in $\mathbb{Z}$ resp.). Then $f(x+r)$ is irreducible iff $f(x)$ is irreducible in $\mathbb{Q}[x]$ (or in $\mathbb{Z}[x]$ resp.).
2. Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ and $a_0 \neq 0$. Define the reciprocal polynomial $\bar{f}(x) = x^n f(\frac{1}{x}) \in \mathbb{Z}[x]$. Show that $\overline{\bar{f}} = f$.
3. Let $f(x), g(x) \in \mathbb{Z}[x]$ with nonzero constant terms. Show that $\overline{fg} = \bar{f}\bar{g}$.
4. Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ where $a_0 \neq 0$ and $n > 1$. Then f is irreducible in $\mathbb{Z}[x]$ iff $\bar{f}$ is irreducible in $\mathbb{Z}[x]$.
5. Show $f(x) = x^4 - 10x^2 + 1$ is irreducible in $\mathbb{Z}[x]$ by first considering $g(x) = f(x+1)$ and then considering $\overline{g(x)}$.
6. Let $h(x) \in \mathbb{Z}[x]$ be irreducible and consider $f(x) = xh(x)$. Show that $f(x)$ is reducible even if $\bar{f}$ is irreducible.